11b · Self-selected ad exposure, on real data — the Criteo experiment (IV · CausalPy)

July 12, 2026

Runs in the `legacy` environment (pymc<6): `make env-legacy`, kernel `cmp-legacy`. First run downloads the 311 MB Criteo file into `~/.cache/cmp` (one-time; a seeded subsample is cached after that).

The business decision. Notebook 11's story, at industrial scale and with nothing simulated: **13.98 million real users** from Criteo's advertising incrementality tests (Diemert et al. 2018). Users who *saw* an ad visit the advertiser at hugely elevated rates — but the ad system *chose* whom to show ads to (bids, browsing, auctions), so exposure is **self-selected** and the naive comparison credits the ad with intent that was already there. We want the true causal effect of exposure, to set the **budget cap per exposed user**.

0.0.1 Why this dataset is a gift for IV

Criteo's incrementality tests randomize **eligibility**: a random 15% of users are locked out of the campaign entirely. That randomized `treatment` flag is a textbook **encouragement instrument** for the self-selected `exposure`:

- **Relevance** — eligibility moves exposure (3.6% of eligible users end up exposed) — *testable*, first-stage F below;
- **Exclusion** — being *silently eligible* does nothing to your visiting behaviour unless an ad actually reaches you — *untestable*, stress-tested in Step 6;
- **Exogeneity** — eligibility is randomized by the test infrastructure — *holds by design*;
- **Monotonicity, for free**: ineligible users are **never** exposed (one-sided noncompliance), so "defiers" are impossible *by construction*, not by assumption — and the LATE specializes to the **effect on the exposed** (ATT), exactly the population a budget cap is about.

0.0.2 What replaces the simulator's planted truth

Two real-data anchors. **(a)** At n = 13.98M the design-based **Wald estimate** (intent-to-treat ÷ first stage) is essentially free of sampling noise — we compute it once on the full file and grade every subsample estimate against it. **(b)** The features let us *photograph the confounding itself*: f0–f11 are balanced across the randomized arms but sharply imbalanced across exposure — the self-selection nb11 could only simulate, measured.

```
FAST=True  subsample=30,000 rows  Bayesian fit on 30,000 rows
```

0.1 2 · Load the real data (what replaces "simulate a ground truth")

`cmp.data.load_criteo_uplift()` downloads the full file once, then caches a **seeded, balanced subsample** — `N_ARM` rows from each randomized arm. Balancing the 85/15 arms is legitimate *because* the sampling key is the randomized flag itself (both the ITT and the first stage are conditional on `treatment`, so selecting on it biases nothing) and roughly doubles precision per row. The **data card**:

fact	value
population	13,979,592 users, assembled from several incrementality tests
randomization	`treatment` = eligible to be targeted (85%) vs locked out (15%)
self-selection	`exposure` = actually saw an ad: **3.6% of eligible, 0.0% of ineligible**
outcomes	`visit` (4.7% base) — primary; `conversion` (0.29%) — secondary, rarer & noisier
features	f0–f11, anonymized dense floats (used for balance checks, not needed for IV)
full-data anchor	Wald LATE on visit **+28.7 pp**, on conversion **+3.2 pp** (computed below)

Provenance check (cheap hygiene). The card's base rates are the published ones: Criteo's dataset release reports 13,979,592 rows, an 85/15 eligibility split, and ~4.7% visit / ~0.29% conversion base rates (Diemert et al. 2018 — full citation in §7), matching what the loader serves. So we are analyzing the dataset the paper describes, not a truncated or corrupted download — reconcile any loader against its source paper before believing anything downstream. The full-file anchor below remains the primary yardstick; this row is provenance, not validation.

The one modelling accommodation, inherited from nb11 and disclosed the same way: outcome and exposure are 0/1, and CausalPy's joint-MvNormal IV treats both equations as **linear-probability** (Gaussian) — fine for the point estimate and the error correlation rho, so those are what we read.

```
subsample: 30,000 rows (15,000 eligible / 15,000 locked out)
P(exposed | eligible) = 0.0372   P(exposed | locked out) = 0.0000
  -> one-sided noncompliance, first-stage F = 580
visit rate: 0.0344 locked out vs 0.0497 eligible (ITT +1.53 pp)
full-data anchor (n=13.98M): ITT +1.034 pp, first stage 0.0360,
  Wald LATE +28.7 pp on visit
```

	f0	f1	f6	treatment	exposure	visit	conversion
0	22.030001	10.06	-7.30	0.0	0.0	0.0	0.0
1	23.740000	10.06	-5.12	0.0	0.0	0.0	0.0
2	12.620000	10.06	0.29	1.0	0.0	0.0	0.0
3	25.770000	10.06	-5.58	0.0	0.0	0.0	0.0
4	23.370001	10.06	-4.60	1.0	0.0	0.0	0.0

0.2 3 · Identify — a real instrument, and the confounding made visible

Exposure is endogenous: the ad system bids more for users who browse more, and browsing predicts visiting with or without ads. The instrument's three conditions are laid out in the header; before photographing the confounding, here is the estimand in symbols — because "the LATE specializes to the ATT" is this notebook's central identification claim, and it deserves a derivation, not a bullet.

0.2.1 The estimand: ITT, first stage, Wald — and why Wald = LATE = ATT here

Notation as in notebook 11 — Z the instrument, X the endogenous exposure, Y the outcome. (The print-outs' "Z->D" arrow is the same first stage: D, for "dose", is the econometrics alias for our X.) With $Z_i \in \{0,1\}$ the randomized eligibility, $X_i \in \{0,1\}$ actual exposure, $Y_i \in \{0,1\}$ a visit, potential outcomes $Y_i(x)$ and potential exposures $X_i(z)$, the three observed quantities are

$$\text{ITT} \;=\; \mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0], \qquad \pi \;=\; \mathbb{E}[X \mid Z{=}1] - \mathbb{E}[X \mid Z{=}0], \qquad \hat{\beta}_{\text{Wald}} \;=\; \frac{\text{ITT}}{\pi}.$$

Why the ratio is causal: write $Y_i = Y_i(0) + X_i\,(Y_i(1) - Y_i(0))$ and take expectations given Z. Randomization (exogeneity) plus exclusion make $\mathbb{E}[Y(0) \mid Z{=}1] = \mathbb{E}[Y(0) \mid Z{=}0]$, so the $Y(0)$ terms cancel from the ITT and only instrument-driven exposure survives:

$$\text{ITT} \;=\; \mathbb{E}\big[(Y(1) - Y(0))\,X \mid Z{=}1\big] \;-\; \mathbb{E}\big[(Y(1) - Y(0))\,X \mid Z{=}0\big].$$

In this design the second term is exactly **zero**: locked-out users are never exposed — $P(X{=}1 \mid Z{=}0) = 0$, printed as 0.0000 in step 2, a fact of the serving infrastructure, not a sample accident. So, with $\pi = P(X{=}1 \mid Z{=}1)$,

$$\hat{\beta}_{\text{Wald}} \;=\; \frac{\mathbb{E}\big[(Y(1) - Y(0))\,X \mid Z{=}1\big]}{P(X{=}1 \mid Z{=}1)} \;=\; \mathbb{E}\big[\,Y(1) - Y(0) \mid X{=}1\,\big] \;=\; \text{the ATT of the exposed}$$

(Bloom 1984) — the conditioning on $Z{=}1$ drops because *every* exposed user is an eligible user. In the general two-sided case this ratio is Imbens–Angrist's complier LATE and needs monotonicity (no defiers) as an *assumption*; one-sided noncompliance delivers it by construction, and the "compliers" are literally the exposed. On the full file the pieces are ITT $\approx +1.03$ pp and $\pi \approx 0.0360$, giving $\hat{\beta}_{\text{Wald}} \approx +28.7$ pp — the anchor every subsample estimate below is graded against.

0.2.2 The confounding, photographed

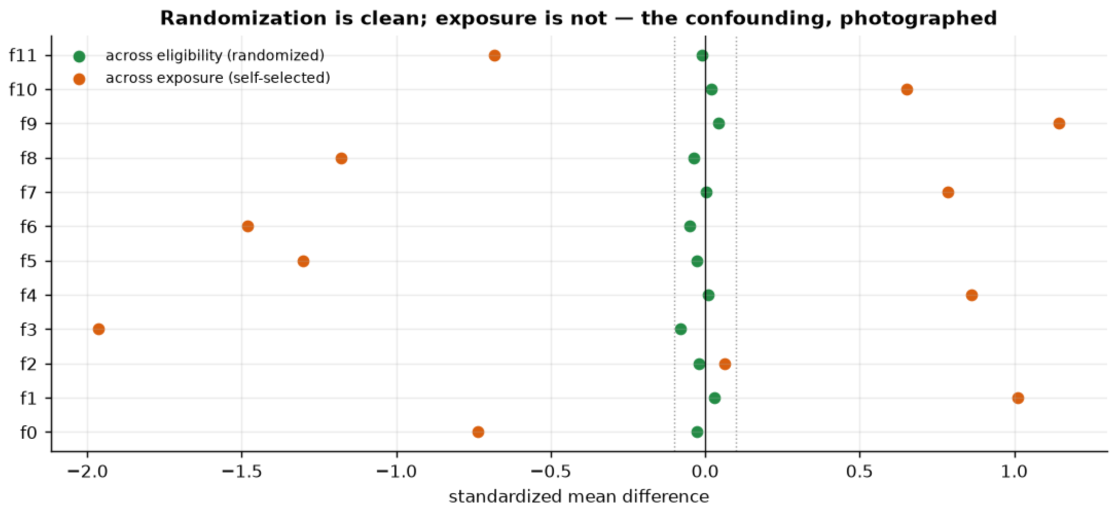
And here is the part a simulator can never show. If randomization worked, the twelve user features must be **flat across the arms** (any imbalance would impeach the instrument's exogeneity). If exposure is truly self-selected, the same features must be **skewed across exposure**. Both facts are checkable — one chart, two stories:

- across **eligibility** (randomized): standardized mean differences ~ 0 -> the instrument is clean;

- across **exposure** (chosen by the auction): large SMDs -> the exposed *were different people before any ad effect* — the very confounder nb11 called `engagement`, photographed in real data.

This is also why no regression on f0–f11 could rescue the naive estimate: these are only the *observable* shadows of targeting; the auction also used signals not in the file (the unmeasured confounder), which is precisely the case where notebook 05's adjustment toolkit surrenders and IV takes over.

```
max |SMD| across eligibility: 0.080  (randomization holds)
max |SMD| across exposure:    1.963  (the exposed were different people to begin
with)
```



0.3 4 · Estimate — OLS, the Wald ladder, and Bayesian IV

Same ladder as nb11, every rung now a real number:

- **OLS / naive**: visit rate of exposed minus unexposed — the number an attribution dashboard would report, inflated by targeting;
- **Reduced form** (eligibility -> visit) and **first stage** (eligibility -> exposure): both clean, both tiny — the encouragement is randomized but weak *per user*, which is why their **ratio** (Wald) is the honest per-exposure effect;
- **Bayesian IV** (CausalPy): the same ratio estimated jointly, with the endogeneity rho estimated rather than assumed away, and weakly-informative priors (nb11's gotcha: CausalPy's default 2SLS-centred `sigma=1` priors would produce an overconfident band). The fit uses `N_FIT` rows — NUTS cost scales with n, and Step 5 grades this subsample posterior against the 13.98M-row anchor to show what the smaller n costs us.

The third rung's model, in symbols — the same joint Gaussian as nb11's, with X = exposure and Y = visit:

$$\begin{pmatrix} X_i \\ Y_i \end{pmatrix} \sim \mathcal{N}_2\left(\begin{pmatrix} b_0 + \pi Z_i \\ \beta_0 + \beta X_i \end{pmatrix},\ \Sigma\right), \qquad \Sigma = \begin{pmatrix} \sigma_X^2 & \rho\,\sigma_X\sigma_Y \\ \rho\,\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix},$$

where β is the causal effect the budget cap needs and the off-diagonal ρ **is** the endogeneity: the correlation between the exposure-equation and visit-equation errors that hidden intent induces and that the naive comparison ignores. The model *estimates* ρ instead of assuming it zero; a positive ρ is the self-selection signature. Two reading notes: ρ is small here even at full fit size — visits are rare events, so even strong targeting induces only a modest error *correlation* — and a small subsample may not resolve its sign at all. The naive-vs-IV gap, not ρ's magnitude, is where the bias shows. Priors: $b_0, \pi, \beta_0, \beta \sim \mathcal{N}(0, 50^2)$ — that is the `sigmas: [50, 50]` passed in the code, the weakly-informative override of CausalPy 0.8.1's default priors, which centre on the 2SLS point with $\sigma = 1$ and yield a band roughly 3x too narrow (nb11 demonstrates that band excluding its planted truth; same model, same library version here, so the gotcha transfers verbatim) — and $\Sigma \sim \text{LKJCholeskyCov}(\eta = 2)$.

One upgrade on the linear-probability disclosure: because Z and X are both binary, both **conditional means** above are *saturated* — a linear function of a binary regressor can match $\mathbb{E}[X \mid Z]$ and $\mathbb{E}[Y \mid X]$ exactly — so the LPM approximation lives entirely in the Gaussian *likelihood*, not in the mean structure. That is precisely why the point estimate and ρ are the trustworthy read-outs, and the equation-level posterior widths are the thing not to over-read.

```
OLS / naive (exposed vs not)   +37.2 pp
Reduced form (Z->Y) +1.53 pp  ÷  first stage (Z->D) 0.0372  =  Wald +41.2 pp
Bayesian IV (30,000 rows)    +40.6 pp   90% credible interval [+31.0, +50.9] pp
full-data anchor (13.98M)      +28.7 pp
IV convergence: max r-hat 1.020 - min ESS 152 - divergences 0
estimated endogeneity rho = -0.02 [90% -0.09, +0.04] - positive rho is the self-
selection: the same hidden leanings push a user into both exposure and visiting
```

0.4 5 · Validate — against 13.98 million rows

No planted truth, but a better yardstick than most real projects ever get: the full-file **Wald anchor**, whose own sampling noise at n = 13.98M is negligible for our purposes. Three checks, plus the star exhibit:

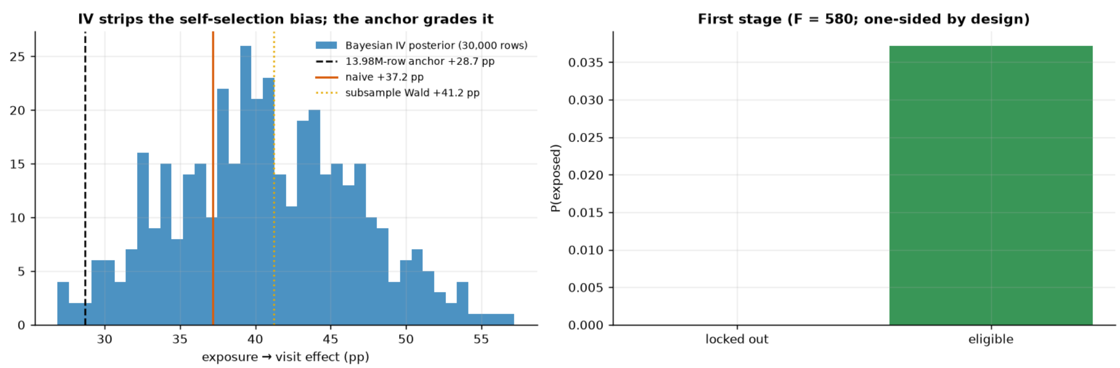
1. **Anchor recovery.** The subsample Bayesian IV's 90% interval should cover the anchor (+28.7 pp), and its point should sit within a couple of pp — that is the real-data analogue of nb11's "recover the €15." (Expect the pass at the full working size; on a small FAST slice the Wald ratio's SE is several pp, so the interval can sit an SE or two off the anchor — subsampling noise that check 3 prices, not a failed identification.)
2. **Internal consistency.** Wald ~ Bayesian IV on the same rows (they are the same identification strategy, estimated two ways); a gap would flag a modelling artefact, not a causal discovery.
3. **Disjoint-subsample stability** — nb11's multi-seed loop, real-data edition: we can't refit on fresh *worlds*, but 13.98M rows let us refit on fresh *samples*. Split the working subsample into disjoint folds, compute each fold's Wald +/- normal CI, and check spread and anchor coverage.
4. **The OLS − IV gap *is* the measured self-selection bias**: the euros an attribution dashboard would over-credit to the ad. On the full file the gap is deterministic: naive ~ +37.9 pp vs the +28.7 pp anchor — the dashboard number runs ~32% too big, on 13.98 million real users. The print below re-measures the same gap on the working subsample, where it also carries the subsample's Wald noise (priced by the fold check).

The fold CIs, derived (and the F-statistic's real job). Each fold's interval is a **delta-method** CI for a ratio. Writing $\hat{\beta} = \widehat{\mathrm{ITT}}/\hat{\pi}$ and propagating both numerator and denominator noise,

$$\mathrm{Var}\,(\hat{\beta}) \;\approx\; \frac{1}{\hat{\pi}^2}\,\mathrm{Var}\,(\widehat{\mathrm{ITT}}) \;+\; \frac{\hat{\beta}^2}{\hat{\pi}^2}\,\mathrm{Var}\,(\hat{\pi}) \;-\; \frac{2\hat{\beta}}{\hat{\pi}^2}\,\mathrm{Cov}\,(\widehat{\mathrm{ITT}},\,\hat{\pi}).$$

The folds keep only the first term, $\mathrm{SE}(\hat{\beta}) \approx \mathrm{SE}(\widehat{\mathrm{ITT}})/\hat{\pi}$. The licence to drop the other two is the strong first stage: the *relative* noise of $\hat{\pi}$ is $1/\sqrt{F}$ (single-regressor first stage, so $F = t^2$ and $\mathrm{SE}(\hat{\pi})/\hat{\pi} = 1/\sqrt{F}$) — a few percent here, versus double-digit relative noise in a fold-level ITT — so the dropped terms move the SE by only a few percent. The code prints the full three-term SE next to the shortcut, so the approximation is audited rather than asserted. This is what a large F actually buys: permission to treat the ratio's denominator as (nearly) known.

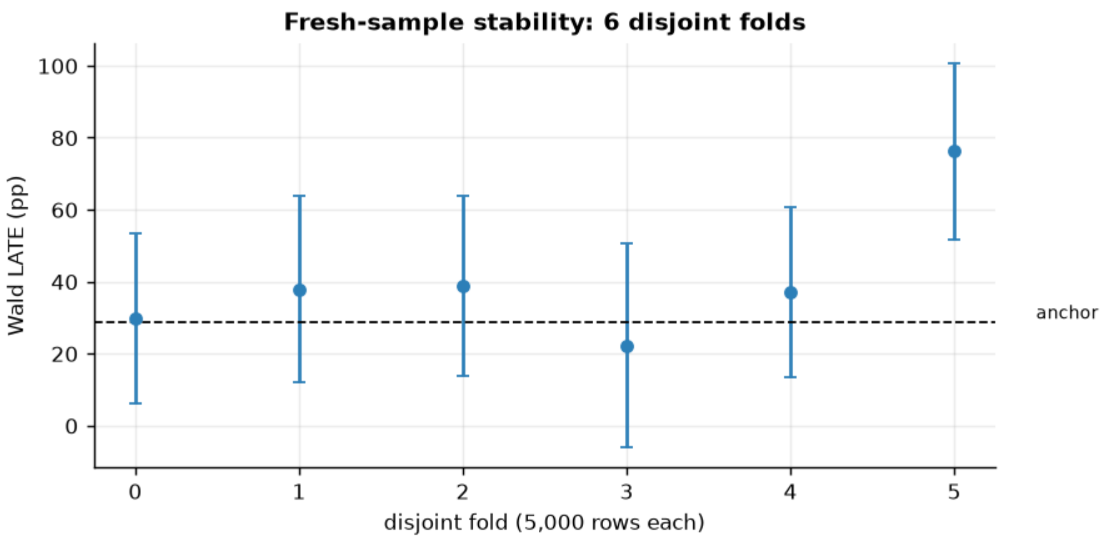
```
anchor recovery: 90% interval [+31.0, +50.9] pp MISSES the 13.98M-row anchor
(+28.7 pp)
internal consistency: subsample Wald +41.2 vs Bayesian +40.6 pp
self-selection bias, measured against the anchor: naive +37.2 -> causal +28.7 pp
(+8.5 pp of pure targeting, ~29% inflation; this slice's own IV: +40.6 pp)
```



How to read this. *Left* — four vertical markers tell the story. The **orange** line is the naive exposed-vs-unexposed gap: the attribution-dashboard number, the one everything else should sit below. The **blue** histogram is the Bayesian IV posterior for the same causal question; the **gold dotted** line is the subsample Wald (same identification, frequentist arithmetic); the **dashed black** line is the 13.98M-row anchor (+28.7 pp). The horizontal distance from the orange line down to the anchor *is* the self-selection premium, measured: on the full file ~9 pp of the ~38 pp naive "effect" — roughly a third — is intent the exposed users already had. The printed checks grade the rest: the 90% interval covering the anchor is check 1, and posterior mean ~ Wald is check 2 (a gap there would flag a modelling artefact, not a causal discovery). One honesty note before over-reading the blue centre: with a 3.6% exposure rate in the denominator, subsample Wald ratios are genuinely noisy, so on a small slice the posterior can drift an SE or two from the anchor — that drift is sampling noise, priced explicitly by the fold chart below, not a broken instrument. *Right* — the first stage is the design itself, not a diagnostic that happened to pass: the locked-out bar is exactly zero (one-sided noncompliance — the fact that turned LATE into ATT in §3) and the eligible bar (~3.6%) is the entire lever the instrument pulls. $F \gg 10$ is the licence for dividing by it; with a weak first stage the left panel would be meaningless.

```
fold Walds: mean +40.4 pp, sd 18.7 pp (anchor +28.7); 90% CIs cover the anchor
in 83% of folds
the spread is the honest price of subsampling a rare exposure (3.6%) - and why
the anchor was computed on all 13.98M rows
```

```
delta-method audit: shortcut SE 6.23 pp vs full three-term SE 6.04 pp (+3%);
first-stage relative noise 1/sqrt(F) = 4.2% - dropping it is disclosed, not
hidden, and errs slightly conservative here
```



0.5 6 · Decide, in euros — the budget cap, and the exclusion stress test

The cap logic of nb11, with real numbers: an exposed user visits with ~ **+29 pp** extra probability (the 13.98M-row anchor: +28.7 pp), so at an assumed **€1 value per incremental visit** an exposure is worth ~ €0.29 — and the *naive* number (full-file exposed-vs-unexposed gap, ~ +37.9 pp) would have priced it at ~ €0.38. The decision rule: **BUY exposure while P(value x LATE > cost) >= 0.9**; sweeping the cost turns that into the **maximum viable price per exposed user**. Both business inputs are assumptions and get swept: the €1 visit-value scales the cap linearly, and the secondary `conversion` outcome gives an independent cap for a margin-per-conversion framing.

The exclusion stress test. The lockout is silent, but eligibility could still leak around the `exposure` flag — e.g. eligible users win *other* auctions differently, or "exposed" mislabels a below-the-fold impression. As in nb11: give eligibility a hypothetical direct effect delta on visits, correct the reduced form, and find the delta that would flip the BUY.

The rule, the cap, and the stress test, in symbols. With v the value per incremental visit, c the cost per exposed user, and $\beta^{(s)}$ the posterior draws of the exposure effect, the decision rule and the two prices the sweep reads off are

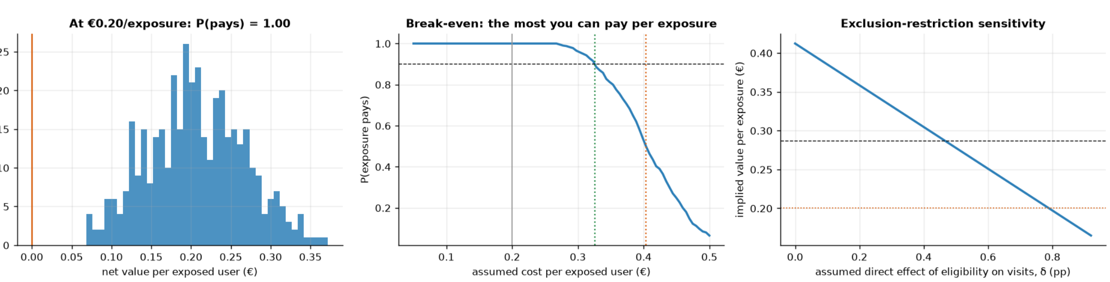
$$\text{BUY while } P(v\,\beta > c) \geq 0.9; \qquad c^* = v\,q_{0.10}(\beta), \qquad c_{50} = v\,q_{0.50}(\beta),$$

where $q_p(\beta)$ is the posterior p-quantile. Sweeping c against a fixed posterior just walks its quantile function — $P(v\beta > c) \geq 0.9 \iff c \leq v\, q_{0.10}(\beta)$ — so the budget cap c^* *is* a posterior quantile denominated in euros, c_{50} is the coin-flip price, and the cap a dashboard would set is $v\,\hat{\beta}_{\text{naive}}$. For the exclusion stress: if eligibility leaks a direct effect δ (in visit probability) around exposure, the honest reduced form is ITT $- \delta$, so the corrected effect and the flip threshold are

$$\beta(\delta) = \frac{\text{ITT} - \delta}{\pi}; \qquad v\,\beta(\delta) > c \iff \delta < \delta^* = \text{ITT} - \frac{c\,\pi}{v}.$$

The flip threshold δ^* has a closed form — no grid search needed — and the code below prints it next to the grid scan of the third panel, so the two agree in front of you (up to the grid's resolution). Note $\beta(\delta)$ re-solves the ratio at point estimates: carry the posterior interval's width alongside it when briefing the cap, as in nb11.

```
at €0.20/exposed user and €1.00/visit: net €+0.21, P(pays) = 1.00
  -> BUY (to the exposed margin)
budget cap: clears the 90% bar up to €0.33/exposed user; a coin flip at €0.40
  - the naive number would have set the cap at €0.37 against the
  anchor's causal €0.29: 29% too high
conversion framing (secondary): full-data anchor LATE +3.20 pp
  -> at €40 margin/conversion an exposure is worth ~€1.28; the subsample
  pins it only to +1.79 pp (90% CI [-0.77, +4.36] pp
  -> €-0.31 to €+1.74) - at a 0.29% base rate this rung is honestly
  noisy; re-derive with the firm's own margin
exclusion stress: the BUY at €0.20 survives until eligibility's direct effect
  exceeds delta ~ 0.80 pp - 52% of the whole reduced form (+1.53 pp)
  would have to bypass exposure
  closed form agrees: delta* = ITT - c·pi/v = 0.79 pp (grid scan: 0.80 pp)
campaign scale: the naive cap embeds €0.08/exposure of pure targeting premium
  (naive €0.37 vs causal anchor €0.29); at 1,000,000 exposures/quarter
  that is ~ €84,660/quarter paid for intent the users already had
```



Read-out — the three panels, then the sentence for the CMO. *Left*: the net-value posterior $v\beta - c$ at the assumed €0.20 cost; the printed P(pays) is its mass above the orange zero line, and the BUY/HOLD call is just whether that mass clears 0.9. *Middle*: the sweep is the posterior's quantile function in euros (§6's $c^* = v\,q_{0.10}(\beta)$) — where the curve crosses the dashed 0.9 line is the green budget cap, the orange dotted line is the coin-flip price $v\,q_{0.50}(\beta)$, and the grey line marks today's assumed cost. *Right*: exclusion stress — the implied value falls linearly as the assumed leak

δ grows (§6's $\beta(\delta) = (\mathrm{ITT} - \delta)/\pi$); the printed δ^* is where the line crosses the orange cost line, and the dashed reference is the anchor's ~€0.29.

What you would tell the CMO. "An exposure to the users this auction actually reaches is worth about €0.29 in expected incremental visits (13.98M-row anchor: +28.7 pp at our assumed €1/visit); cap bids at the printed 90%-confidence cap, a few cents below that. The attribution dashboard prices the same exposure at ~ €0.38 (full-file naive gap: +37.9 pp) — about a third of that is intent we already owned, so a dashboard-calibrated cap overpays on *every* exposure: ~ €0.09 each, order of €90k per quarter at a million exposures a quarter (the print above re-computes this at this run's numbers). Three scope lines before this goes into a rate card: it is the ATT of the ~3.6% of users the auction actually reaches — it prices *these* exposures and does not license doubling reach, because the next users the auction would find are, by its own logic, worse prospects; the €1 visit-value and €40 margin are swept assumptions, so finance should re-derive the cap with its own numbers; and the BUY survives an eligibility side-channel only up to the printed δ^* — a leak bigger than that flips the call."

0.6 7 · Caveats

- **ATT of the exposed, not everyone.** One-sided noncompliance makes monotonicity free, but the estimate speaks only for the ~3.6% of users the auction actually reached — the right population for "what is an exposure worth," the wrong one for "what if we doubled reach" (the next users the auction would reach are, by its own logic, worse prospects).
- **Exclusion is the load-bearing untestable.** The lockout is silent, but any eligibility side channel (auction dynamics on other campaigns, mislabelled below-the-fold "exposures") leaks into the estimate; step 6 priced the leak that would flip the call.
- **Linear-probability approximation** — binary outcome and binary exposure in a Gaussian joint model, as in nb11: read the point and rho, not the equation-level posterior widths.
- **Visits != profit.** The €1/visit and €40 margin/conversion are finance inputs, swept but assumed; the conversion-based rung is noisier (0.29% base rate) and its cap should be re-derived at decision time with the firm's own margin.
- **A composite of several tests** — Criteo assembled multiple incrementality tests; the LATE is a blend across campaigns/periods (fine for a benchmark cap, blur it into one campaign's plan with care).
- **Subsampling is disclosed, not hidden**: every number above is grade-able against the full-file anchor, and the fold check shows exactly how much noise a 200k-row view carries.

Cite the data: Diemert, Betlei, Renaudin & Amini (2018), *A Large Scale Benchmark for Uplift Modeling*, AdKDD. Dataset: Criteo AI Lab, research use.